

11. Newman Command Line Execution

In API testing, **Newman** is a powerful tool used to run Postman collections directly from the **command line (CLI)**. It is mainly used for **automation, CI/CD integration, and continuous testing**.

In this project using Fake Store API, Newman helps us execute the entire API test suite (Products, Cart, Users, Authentication) without opening Postman UI.

11.1 What is Newman?

Newman is the **command-line runner for Postman collections**.

☞ It allows you to:

- Run Postman collections from terminal
 - Automate API testing
 - Integrate with CI/CD tools (Jenkins, GitHub Actions)
 - Generate test execution reports
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11.2 Why Newman is used?

We use Newman because:

- It supports **automation testing**
 - It removes dependency on Postman UI
 - It is **CI/CD friendly**
 - It provides **faster execution**
 - It is useful for **regression testing**
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11.3 How Newman works in this project

In our Fake Store API project:

Step 1:

Export Postman collection → `collection.json`

Step 2:

Export environment file → `environment.json`

These files contain:

- API requests
 - URLs
 - Tokens
 - Variables
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Step 3:

Run Newman command in terminal

11.4 Basic Newman Command

```
newman run collection.json
```

☞ This runs the full API collection.

11.5 Running with Environment File

```
newman run collection.json -e environment.json
```

☞ This ensures:

- baseUrl is used dynamically
 - authToken is reused
 - productId/cartId values are shared
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11.6 Important Newman Options

◆ 1. Iteration (Run multiple times)

```
newman run collection.json -n 5
```

☞ Runs collection 5 times.

◆ 2. Delay between requests

```
newman run collection.json --delay-request 1000
```

☞ Adds 1-second delay between APIs.

◆ 3. Export Results

```
newman run collection.json -r cli,json
```

☞ Output formats:

- CLI (terminal output)
 - JSON (for reporting)
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11.7 Real Execution Flow in Project

When we run Newman:

Step 1: Authentication API runs

- Token is generated

Step 2: Product APIs execute

- GET all products
- GET single product
- POST / PUT / PATCH / DELETE

Step 3: Cart APIs execute

- Add to cart
- Update cart
- Delete cart

Step 4: User APIs execute

- Fetch users data

☞ Entire workflow runs automatically in sequence.

11.8 Newman in CI/CD Pipeline

Newman is widely used in DevOps pipelines.

Example flow:

1. Developer pushes code
2. Jenkins triggers build
3. Newman runs API tests
4. Results generated
5. If any test fails → pipeline fails

☞ This ensures **early bug detection**

11.9 Advantages of Newman

- Fully automated execution
 - No GUI required
 - Easy CI/CD integration
 - Fast execution
 - Supports reporting tools
 - Works in any environment (Windows/Linux/Mac)
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11.10 Limitations of Newman

- Requires Node.js installation
 - No visual interface
 - Debugging is harder than Postman UI
 - Needs proper setup for reports
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✓ Final Interview Summary

In this project, I used Newman CLI to execute the entire Fake Store API test suite from the command line. I exported Postman collections and environment files, and executed them using Newman commands. It helped me automate API testing, integrate with CI/CD pipelines, and validate end-to-end workflows like authentication, product management, cart operations, and user validation. This improved efficiency, reduced manual effort, and enabled continuous testing.

